Likelihood-based simulation and inference for typed multi-relational network dynamics

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Many social, economic and political systems are networks of several kinds of entities linked by several kinds of relationships that form and dissolve at rates depending on the surrounding network: friends of friends become friends, states ally with their allies' allies, agents adopt a behavior when enough contacts have adopted it. We present tricl, an open-source simulator and inference engine for such dynamics. A model is a short configuration file that declares entity types, relationship types and rules saying how the attempt rate and the success probability of establishing or terminating a link depend on the numbers of adjacent two-paths of each type. States of entities and well-mixed interactions are expressed with the same rules through links to auxiliary entities. The resulting continuous-time Markov chain is simulated exactly with the direct method over a sum tree of event rates. Because every rule enters the rates explicitly, tricl also computes the exact log-likelihood of an observed event sequence and its gradient with respect to all model parameters, which turns the simulator into a maximum-likelihood estimator, and it generates mean-field equations from the same file. We describe the model class and the algorithms, demonstrate parameter recovery and the mean-field approximation, and apply the method to the formation and dissolution of defense pacts between states from 1816 to 2018, where a common ally raises the rate of pact formation about 58-fold.

I. INTRODUCTION

Networks change. Friendships form and fade, states sign and abandon alliances, people adopt and drop behaviors, pathogens spread over contacts that are themselves avoided once someone is known to be infectious. In each of these examples the rate at which a tie forms or dissolves depends on the surrounding network, and the network usually has more than one kind of node and more than one kind of tie. Models of such *adaptive* or *coevolutionary* networks are a large field in statistical physics [1–3], and stochastic models of network dynamics are a large field in statistics and the social sciences [4–9]. Typically, each model is implemented anew, for one kind of node and one kind of tie, and the simulation code and the statistical code are separate programs, if both exist.

tricl ("triadic closure", after its first use case) takes a different route. A model is a configuration file in a small rule language, and a single program simulates it, computes the likelihood of observed data under it and derives a macroscopic approximation from it. The rule language has one construct: the rate of an event that establishes or terminates a link between two entities is a function of the numbers of *angles*, i.e. two-paths of each type, that connect the two entities. Triadic closure, structural balance, homophily via shared attributes, threshold dynamics and compartmental epidemics are all of this form once states of entities are encoded as links to auxiliary entities (Sec. II). The class is a continuous-time Markov chain on typed multi-relational graphs [10], so it can be simulated exactly [11, 12], and its likelihood for a fully observed event sequence is an explicit sum and integral of event rates.

This paper describes the model class (Sec. II), the simulation algorithm and its tests (Sec. III), likelihood-based inference with analytic gradients (Sec. IV), the automatically generated mean-field approximation (Sec. V), and an application to the formation and dissolution of military alliances between states (Sec. VI). Section VII discusses the relation to other model families and the limitations.

II. MODEL CLASS

A. State

A tricl model has a finite set of *entities*, each of one *entity type*, and a finite set of *relationship types*. The state at time t is a set $G(t)$ of *links* $\ell = (e_1, r, e_3)$, each a directed edge of relationship type r from entity e_1 to entity $e_3 \neq e_1$. A relationship type may be declared *symmetric* (e.g. `knows`) or have a named *inverse* (`follows`/`is followed by`); then every link implies its mirrored link, and the two are established and terminated together. A built-in identity relationship "=" links every entity to itself and never changes; it lets rules refer to an existing link between the two entities, or to a link of one of them to a third entity of a given kind, with the same syntax as to two-paths. Apart from the identity relationship there are no self-links, as in the original implementation: a self-link (e_1, r, e_1) would turn every link (e_1, r', e_3) into a two-path $e_1 \xrightarrow{r} e_1 \xrightarrow{r'} e_3$ of the same type as a two-path through a third entity, so that, e.g., an existing acquaintance would count as a common acquaintance; states of entities are encoded by links to auxiliary entities instead (Sec. II E). The *link type* of ℓ is the triple (type of e_1, r, type of e_3).

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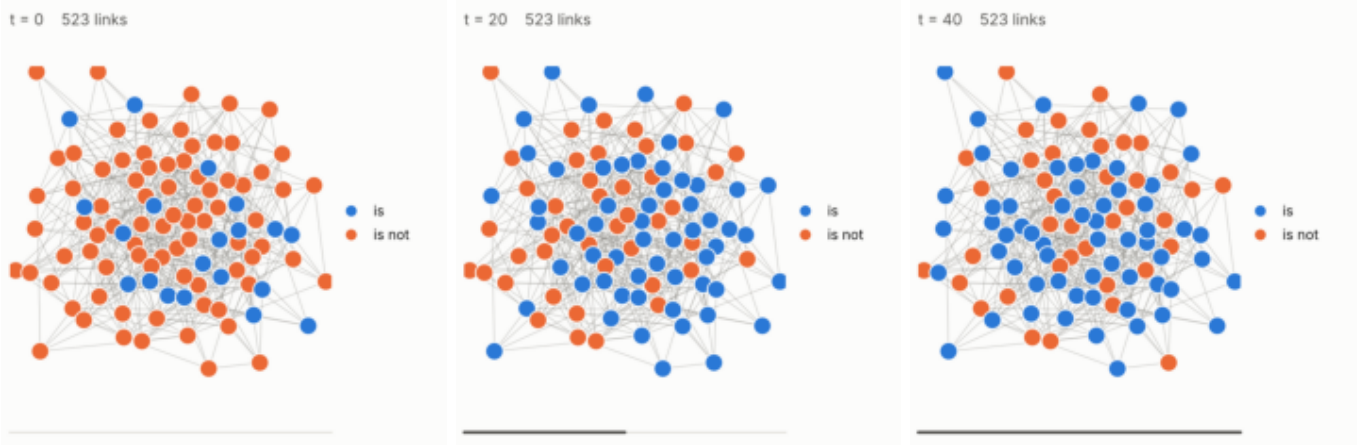


FIG. 1. Three moments of a simulation of a threshold model of social tipping (the `granovetter_helfmann` example of `tricl`, seed 3). The network has 100 agents of three kinds, always active, contingent and never active; contingent agents become active or inactive depending on the fraction of their active neighbors. The state of an agent is a link to an auxiliary entity `active` (relationship types `is` and `is not`), so the threshold rule is a rule on two-paths agent–neighbor–`active`. Node positions come from a force-directed layout of the time-aggregated network and are the same in all panels.

B. Events and rates

An *event* establishes a link that does not exist or terminates one that exists. For a candidate link $\ell = (e_1, r, e_3)$, an *angle* is a two-path $e_1 \xrightarrow{r_{12}} e_2 \xrightarrow{r_{23}} e_3$ in $G(t)$ with r_{12}, r_{23} including the identity relationship; its *angle type* is $(r_{12}, \text{type of } e_2, r_{23})$. For every event type τ (establish or terminate, link type), the configuration file specifies a basic attempt rate $a^0_\tau \geq 0$, basic success *probability units* $b^0_\tau \in \mathbb{R} \cup \{\pm\infty\}$, and for any angle type k an additional attempt rate $a_{\tau k}$ and additional probability units $b_{\tau k}$ contributed by each adjacent angle of that type. With $n_k(\ell, t)$ the number of angles of type k adjacent to ℓ at time t, the event happens at rate

$$\begin{aligned} \lambda_\tau(\ell, t) &= A \cdot f(B), \\ A &= a^0_\tau + \textstyle\sum_k a_{\tau k}\, n_k(\ell, t), \\ B &= b^0_\tau + \textstyle\sum_k b_{\tau k}\, n_k(\ell, t), \end{aligned} \tag{1}$$

i.e. attempts occur at the rate A and succeed with probability $f(B)$. Coefficients may be expressions of named *metaparameters*, which can be overridden on the command line and are the parameters estimated in Sec. IV. Infinite coefficients encode logic: a contribution $-\infty$ to B makes the event impossible, and an infinite attempt rate with positive success probability makes it *immediate*; all pending immediate events are performed, in a uniformly random order, before time advances. Immediate events implement state machines: if an agent acquires a link `is infectious for` to the entity `covid-19`, the rule "terminate `is susceptible to` immediately when the angle `is infectious for`, `covid-19`, = exists" removes its previous state.

C. The sigmoid

The success probability is a sigmoid f of the probability units with two *tail indices* $\sigma_0, \sigma_1 \geq 0$ that set how fast it approaches 0 and 1:

$$\begin{aligned} f(B) &= \tfrac{1}{2} T_{\sigma_0}(-v) + \tfrac{1}{2} - \tfrac{1}{2} T_{\sigma_1}(v), \quad v = \frac{4B}{k(\sigma_0) + k(\sigma_1)}, \\ T_\sigma(x) &= \left(1 + \sigma \ln(1 + e^x)\right)^{-1/\sigma}, \quad T_0(x) = \frac{1}{1 + e^x}, \\ k(\sigma) &= (1 + \sigma \ln 2)^{-1-1/\sigma}, \quad k(0) = \tfrac{1}{2}. \end{aligned} \tag{2}$$

T_σ replaces the exponential tail of the logistic function by the q-exponential $(1 + \sigma y)^{-1/\sigma}$ of $y = \ln(1 + e^x)$ [13]. For $\sigma > 0$ the tail decays like $|B|^{-1/\sigma}$, for $\sigma = 0$ exponentially. Since $k(\sigma) = -2T'_\sigma(0)$, f has slope 1 at $B = 0$ for all tail indices, so that near zero, probability units are differences of probabilities. f is strictly increasing, continuous in the tail indices, and equals expit$(4B)$ for $\sigma_0 = \sigma_1 = 0$. Heavy tails (the default is $\sigma_0 = \sigma_1 = 1$) keep rare events possible when many angles push the probability units far from zero.

D. A model file

The following excerpt is the dynamics of a model of triadic closure with triangle stabilization in three blocks of 100 entities (the `parameters_3blocks` example):

```
relationship types:
  knows: symmetric
dynamics:
  [user, knows, user]:
    establish:
      attempt:
```

```
    basic: 0.001                      # spontaneous
    [~, knows, user, knows, ~]: 0.01 # per common
  success:                            # acquaintance
    basic: .inf                       # always succeed
terminate:
  attempt:
    basic: 10.0                       # fast decay ...
  success:                            # ... hampered
    [~, knows, user, knows, ~]: -0.3 # by triangles
```

The angle type `[~, knows, user, knows, ~]` counts common acquaintances. Further sections of the file declare entity types and numbers, named entities, and initial links, which can be listed, read from csv files, or drawn from random graph models (Erdős–Rényi, block models, random geometric graphs).

E. Expressiveness

Entity states are links to auxiliary "hub" entities: an agent is active if it has the link `is active`. A rule on the angle (agent, `knows`, agent, `is`, `active`) then depends on the number of active neighbors, which yields threshold models [14, 15] (Fig. 1) and contact processes; angles through "=" make rates depend on an entity's own state, which yields compartmental epidemic models on adaptive networks [2], including social distancing as termination of contacts with infectious agents. Structural balance dynamics [16, 17] needs two relationship types `likes` and `dislikes` and four angle rules. Well-mixed interactions are angles through a hub to which all entities are linked. What is outside the class: rates that depend on structures beyond two-paths (e.g. on cycles of length four) and rates that depend on time or on the history of the process. Rules on single links of one endpoint ("legs", which would make rates depend on degrees) are part of the configuration syntax but rejected by the current version.

III. SIMULATION

A. Algorithm

The process is a continuous-time Markov chain with finitely many possible events, so it can be simulated exactly by drawing the waiting time until the next event from an exponential distribution with the total rate and then choosing the event with probability proportional to its rate [11, 12]. The number of possible events grows quadratically with the number of entities, but in sparse networks most of them are spontaneous establishments of links between entities without any adjacent angle, whose rates are equal within an event type. tricl therefore schedules three kinds of events:

- *Individual events*: all terminations of existing links and all establishments with at least one adjacent angle of an influencing type. Their rates are the leaves of a complete binary *sum tree*, whose inner nodes hold the sums of their children. Updating a rate costs $O(\log n)$ for n scheduled events; the root holds the total, and choosing an event with probability proportional to its rate is a descent from the root.

- *Summary events*, one per type of spontaneous establishment, which stand for all pairs of entities of the two types. A summary event's leaf holds the upper bound $a_\tau^0 p_\tau^{\max} N_1 N_3$, with N_1, N_3 the numbers of entities of the two types and $p_\tau^{\max}$ the largest success probability a covered pair can have. When it is chosen, a pair is drawn uniformly; the attempt is discarded if the pair is identical, already linked or scheduled individually, and otherwise succeeds with probability $f(B)/p_\tau^{\max}$. This is thinning in the sense of Lewis and Shedler [18]: proposals arrive at the upper-bound rate and are accepted with the ratio of the true to the bounding rate, which yields exactly the true rate for every pair.

- *Immediate events* (infinite rates) are kept in a separate list. While it is not empty, one of its events is drawn uniformly and performed without advancing time.

After an event on the pair (e_1, e_3), only events whose angle counts change need new rates: those of pairs (e_1', e_3) with a link between e_1' and e_1, and of pairs (e_1, e_3') with a link between e_3 and e_3', for the relationship types that occur in rules. The cost per event is thus $O\big((d_1 + d_3)\log n\big)$ in the degrees d_1, d_3 of the endpoints. Adjacent angles are found by intersecting the sorted link sets of the two endpoints, with a galloping search when one set is much larger than the other, so that the intersection costs time proportional to the smaller set times the logarithm of the larger one; a hub entity linked to all others (the usual encoding of states) therefore does not make every event cost time linear in the number of entities. An earlier version scheduled events by tentative times in an ordered map [19]; the direct method needs two random numbers per event instead of one per rate change and gives the exact total rate at all times, which the likelihood (Sec. IV) needs.

B. Tests

The test suite runs in continuous integration in optimized, debug and sanitizer builds. It checks the sigmoid (81 063 unit checks of values, limits, monotonicity and derivatives), the exact log-likelihood against a closed form for a two-entity model, analytic gradients against finite differences, that replaying a simulated event sequence reproduces the simulation's log-likelihood, and, in debug builds, the consistency of every scheduled rate with a recomputation from the network. Replacing the scheduler left the distribution of all reported statistics

TABLE I. Throughput of tricl on one core of a cloud virtual machine (best of three runs of 2×10^4 events, initialization excluded). Helfmann: threshold model of Fig. 1 with mean degree 10; 3-blocks: the triadic-closure model of Sec. II, mean degree about 100; SIR: epidemic with social distancing on a random geometric graph.

model	agents	events	init. [s]	events/s
Helfmann	100	20 000	0.07	245 400
Helfmann	1 000	20 000	0.14	164 600
Helfmann	10 000	20 000	2.25	62 200
3-blocks (dense)	300	20 000	1.08	2 400
SIR with distancing	1 000	7 155	0.25	33 000

unchanged within sampling error (between 60 and 400 seeds per model; all $|z| < 1.6$).

C. Performance

Table I reports throughput on one core. Sparse models run at 10^4–10^5 events per second, and a hundredfold increase in the number of agents of the threshold model reduces the throughput only about fourfold. In dense models every event changes the rates of hundreds of others, so the throughput is limited by the number of rate updates. Memory use is a few tens of megabytes for these models.

IV. LIKELIHOOD AND ESTIMATION

A. Log-likelihood of an event sequence

Let $(t_i, \ell_i, \tau_i)_{i=1}^m$ be a fully observed sequence of events in an observation window $[0, T]$, starting from a known initial network. Between events the network is constant, so the total rate $\Lambda(t) = \sum_{\text{possible events}} \lambda$ is piecewise constant, and the log-likelihood of the sequence under parameters θ is

$$L(\theta) = \sum_{i:\ \lambda_i < \infty} \log \lambda_{\tau_i}(\ell_i, t_i^-; \theta) - \int_0^T \Lambda(t; \theta)\, dt + \sum_{i:\ \lambda_i = \infty} \log \frac{1}{m_i}, \tag{3}$$

where the last sum runs over immediate events and m_i is the number of immediate events pending when event i happened (their order is uniformly random). The integral is a sum of $\Lambda\times$ waiting time over the inter-event intervals and the final interval up to T. tricl maintains Λ incrementally, recomputes it exactly every 2^{16} events against floating-point drift, and accumulates Eq. (3) during simulation. In *replay mode* (option `--events-in`) it reads an event sequence from a csv file instead of drawing events, applies them in order, checks that each is possible under the model, and reports $L(\theta)$. Replay draws no random numbers.

B. Gradient

By Eq. (1), λ is linear in the attempt-rate coefficients and depends on the probability-unit coefficients through f. With n_k the angle counts of an event,

$$\frac{\partial L}{\partial a_{\tau k}} = \sum_{i:\ \tau_i = \tau} \frac{n_k(\ell_i)}{A_i} - \int_0^T \sum_{\ell} n_k(\ell, t)\, f(B_\ell(t))\, dt,$$

$$\frac{\partial L}{\partial b_{\tau k}} = \sum_{i:\ \tau_i = \tau} \frac{n_k(\ell_i) f'(B_i)}{f(B_i)} - \int_0^T \sum_{\ell} n_k(\ell, t)\, A_\ell(t)\, f'(B_\ell(t))\, dt,$$

with sums over the possible events ℓ of type τ (and $n_k \equiv 1$ for the basic coefficients a_τ^0, b_τ^0). The integrands are "exposures" that tricl maintains like the total rate, so the full gradient costs about as much as the likelihood itself (option `--grad`). Metaparameters enter the coefficients through expressions; the Python driver `tricl_fit.py` maps the gradient to metaparameters by the chain rule, with the Jacobian from finite differences of the parsed coefficients, and maximizes L with L-BFGS-B [20]. Standard errors come from the observed information, the negative Hessian obtained by finite differences of the analytic gradient.

For fixed probability units, L is concave in the attempt rates (a sum of logarithms of linear functions minus a linear function). In the probability units it is not concave in general, because f is sigmoidal; Sec. VI shows a case where the profile likelihood is not unimodal. Two identifiability limits are structural: for an event type without any varying angle count only the product $a_\tau^0 f(b_\tau^0)$ is identified; and an angle type's attempt-rate and probability-

TABLE II. Parameter recovery: true values, mean and standard deviation of the estimates over 40 simulated event sequences, mean estimated standard error, and the share of sequences whose 95% interval (estimate ±1.96 standard errors) contains the true value.

parameter	true	mean est.	s.d. est.	mean s.e.	coverage
A	0.5	0.503	0.022	0.028	1.00
K	0.8	0.812	0.102	0.096	0.90
D	0.7	0.701	0.045	0.040	0.95
B_K	-0.4	-0.399	0.037	0.039	0.97

unit coefficients are separately identified only if its count varies enough across events.

C. Parameter recovery

We simulated the three-entity model of the test suite (triangle closure: establishment attempts at rate $A + K\,n$ for n common neighbors, termination attempts at rate D with probability units $B_0 + B_K n$) for 400 time units and 40 seeds and re-estimated A, K, D and B_K from each event sequence, starting away from the true values (about 1040 events per sequence). Table II shows estimates centered on the true values, estimated standard errors of the size of the spread across seeds (within about 30%), and coverage of the nominal 95% intervals between 0.90 and 1.00, consistent with 95% given 40 sequences.

V. MEAN-FIELD APPROXIMATION

Because a model is a finite set of typed rules, its mean-field equations can be generated mechanically. `tricl_macro.py` reads the model structure from `tricl --dump-model` and takes as state variables the numbers n_L of links of every link type L. For a candidate link of type L between entities of types α and γ, the number of adjacent angles of type (r_{12}, β, r_{23}) is approximated by a Poisson variable with mean $N_\beta\,\rho(\alpha, r_{12}, \beta)\,\rho(\beta, r_{23}, \gamma)$, independently across angle types, where ρ is the current density of the respective link type. Averaging the rates of Eq. (1) over these distributions gives

$$\frac{dn_L}{dt} = (\text{unlinked pairs of type } L) \cdot \langle \lambda_L^{\text{est}} \rangle - n_L \cdot \langle \lambda_L^{\text{term}} \rangle, \tag{4}$$

with the necessary corrections for symmetric and inverse relationship types and for immediate events. This is the pair-level mean-field closure of the adaptive-network literature [2, 21]; for models whose rates do not depend on angles it is the exact equation for the expected counts, and the law of large numbers for density-dependent Markov chains [22] then makes it the limit for large systems.

Figure 2 compares the approximation with simulations. For linear state switching (no angle influences) it agrees within sampling error. For an infection process on a random geometric graph it overestimates the prevalence substantially, because infections cluster in space and infected agents' neighbors are more often already infected than the closure assumes; for the threshold model it underestimates the number of active agents. The generated equations thus provide a baseline whose deviation from the simulation measures the effect of link correlations, and a starting point for higher-order closures [21, 23].

VI. APPLICATION: MILITARY ALLIANCES 1816–2018

A. Data and model

The Alliance Treaty Obligations and Provisions project [24] records for every pair of states and year whether an alliance with a defense pledge was in effect; we use version 5.1 as distributed with the R package `peacesciencer` [25], together with the Correlates of War list of state system members (version 2016). Every maximal run of consecutive years in which a pair has a defense pledge becomes one interval of a symmetric link `defense`, established at the start of its first year and terminated at the start of the year after its last one. Membership of a state in the international system is a link `[state, is in, system]` to a single hub entity. This gives 217 states, 3 000 pact intervals (1 707 still in effect in 2018) and 4 514 events (Fig. 3, left). The model has six meta-parameters:

```
[state, defense, state]:
  establish:
    attempt:
      [~, is in, system, contains, ~]: a0  # per pair
      [~, defense, state, defense, ~]: a1  # per ally
    success: inf
  terminate:
    attempt:
      base: d0
    success:
      tails: 0
      [~, defense, state, defense, ~]: -b1 # per ally
[state, is in, system]: rates "entry", "exit"
```

Formation thus needs both states to be members (the angle through the hub), and its rate grows linearly in the number of common allies; dissolution attempts succeed with probability $\text{expit}(-4b_1 k)$ for k common allies.

Two features of the data need a convention. Pledges are recorded per year, so events of the same year are simultaneous; we order them as terminations, exits, entries, formations, and formations by state codes. And a multilateral treaty appears as many dyadic events in one year, so that the later ones already see the earlier ones as common allies. With a random order of the events within a year, the estimates change by the amounts dis-

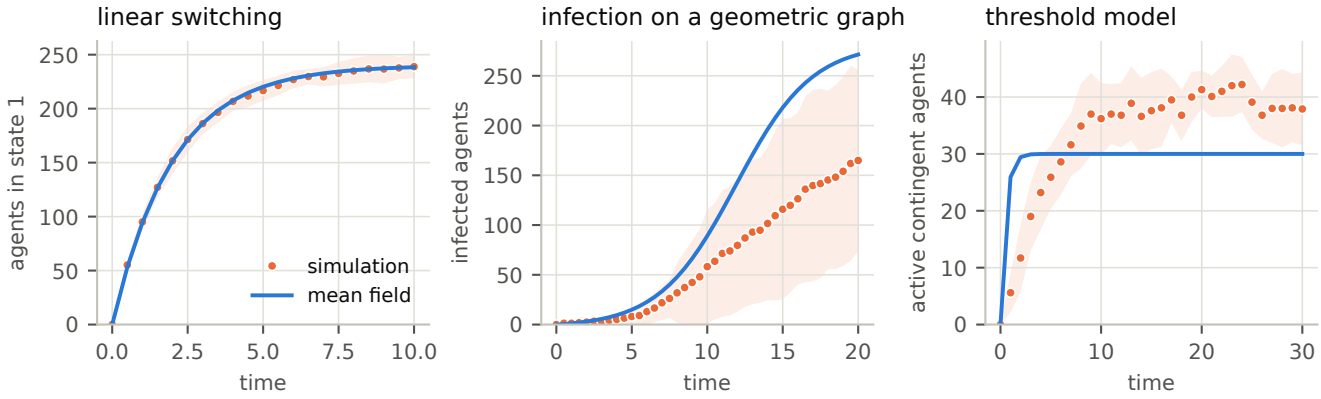


FIG. 2. Automatically generated mean-field approximation (lines) and simulations (mean and ±1 standard deviation over seeds, markers and bands). Left: 400 agents switching independently between two states (20 seeds). Middle: infection with recovery on a random geometric graph of 1000 agents (10 seeds). Right: number of active contingent agents in the threshold model of Fig. 1 (10 seeds).

cussed below.

B. Results

Table III reports the maximum-likelihood estimates. A pair of member states without a common ally forms a pact at a rate of 1.78×10^{-4} per year; each common ally adds 0.0102, so that one common ally raises the formation rate about 58-fold. The dissolution hazard is $d_0 \operatorname{expit}(-4b_1 k)$: 0.054 per year without common allies (a mean pact duration of 18 years), 0.035 with five and 0.0022 with 26, the size of NATO in 2004 (Fig. 3, right). Both effects are strongly supported: removing the ally-of-ally term from the formation rule lowers the maximal log-likelihood by 9 314, removing the stabilization term by 776. With a random order of the simultaneous events, a_0 changes by 28%, a_1 by 4% and b_1 by 1%, so the formation effect partly reflects treaty-level events, which a model with an explicit treaty entity type could separate.

The profile log-likelihood of b_1 (Fig. 3, middle) is not unimodal. Below a local minimum at small negative values it rises again towards a plateau as $b_1 \to -\infty$, where any common ally makes every dissolution attempt succeed; the plateau lies about 891 log-likelihood units below the maximum. An optimiser started at negative values drifts towards the plateau, which illustrates the non-concavity in the probability units noted in Sec. IV; the generated model therefore starts the fit at a positive value. A likelihood evaluation of the 4 514 events takes about 0.06 seconds, a complete fit with standard errors about a minute.

VII. DISCUSSION

a. Relation to other model families. Stochastic actor-oriented models [4–6] describe network change by actors who reconsider one outgoing tie at a time according to an objective function, and are estimated from panel data by the method of moments or by likelihood-based methods with data augmentation [9]. Relational event models [7] and dynamic network actor models [8] model sequences of time-stamped interactions with rates that depend on statistics of the past. tricl is tie-oriented rather than actor-oriented: each potential link has its own establishment and termination rates, which depend on the current two-path structure. It shares with relational event models the exact likelihood of a fully observed event sequence, and with the adaptive-network literature [1] the coupling of states and ties, which it expresses within the same rule language through hub entities. Its distinctive features are typed multi-relational networks with symmetric and inverse relationship types, models as configuration files, immediate events for state logic, and the derivation of simulation, likelihood and mean-field equations from the same file. The interval output of a run is a temporal network [26] in the edge-list form that temporal-network software reads, and a mapping to the Resource Description Framework (entity types as classes, relationship types as properties, symmetric and inverse types as their OWL counterparts, link intervals as annotated triples) lets tricl import typed graphs from knowledge bases and export trajectories to them.

b. Limitations. Estimation requires a complete event sequence, including the timing of events. Most network data are snapshots or panels; for them the likelihood is an integral over unobserved paths, which needs simulation-based methods such as data augmentation or the method of simulated moments, both of which the fast

TABLE III. Maximum-likelihood estimates for the alliance model (standard errors from the observed information). Rates per year.

parameter	meaning	estimate	s.e.
a_0	formation rate per pair of member states	1.78×10^{-4}	8.2×10^{-6}
a_1	additional formation rate per common ally	0.0102	2.0×10^{-4}
d_0	dissolution attempt rate	0.108	0.0048
b_1	probability units per common ally (dissolution)	0.0371	9.2×10^{-4}
entry	entry rate of states	0.00806	5.4×10^{-4}
exit	exit rate of states	0.00315	4.4×10^{-4}
log-likelihood			−20 841.2

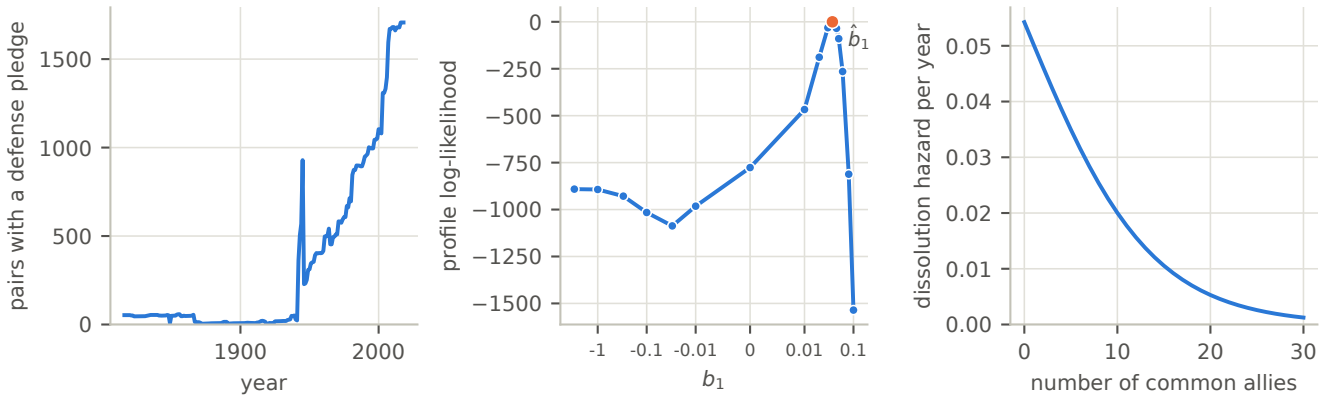


FIG. 3. Defense pacts between states. Left: number of pairs of states with a defense pledge in effect, per year. Middle: profile log-likelihood of the stabilization parameter b_1 (all other parameters re-estimated), relative to its maximum (symmetric-log horizontal axis). Right: estimated dissolution hazard as a function of the number of common allies.

simulator makes feasible but which are not implemented yet. The rule language covers two-path dependence only, and the coefficients do not depend on time; rules on single links of one endpoint (degree dependence) are part of the syntax but not yet supported. The mean-field closure ignores correlations between links (Sec. V). In the application, the yearly resolution and the dyadic coding of multilateral treaties make the formation effect partly a property of treaties rather than of pairs.

c. Outlook. Next steps are higher-order (pair and triple) closures generated in the same way, estimation from snapshot data, rules on single links, and applications with event-level data, e.g. online follow networks and scientific collaboration, where the typed relationships (co-authorship, citation, shared affiliation) are central.

CODE AND DATA AVAILABILITY

tricl is free software under the GNU General Public License v3 at `https://github.com/pik-gane/tricl`. The scripts in its directory `paper/` regenerate every figure, table and number of this paper from the code and from public data.

USE OF AI TOOLS

Tools. This work used Anthropic's Claude Code (cloud sessions, September–November 2026) with the models [MODELS], including parallel sub-agents of the same models.

Substantive use. Under the author's direction, the AI system wrote most of the changes in the current version of tricl: the bug fixes in the likelihood and the rate bookkeeping, the performance work and the sum-tree scheduler, the new form of the sigmoid (after the author's specification), the replay mode with analytic gradients, and the Python tools for estimation, parameter recovery, the mean-field approximation, RDF conversion, movies and the alliance data, together with their tests. It ran the simulations, fits and benchmarks reported here, wrote the scripts producing the figures and tables, searched and checked the literature, and drafted the text of this paper.

Assistive use. Copy-editing, consistency checks between text, code and generated numbers, and checking of bibliographic data against Crossref.

Direction and verification. The author designed the model class and wrote the original simulator, commissioned and approved a written development plan and its order of work, specified the sigmoid, chose the application and the analyses, and reviewed the changes and the text. The code is verified by an automated test suite run in continuous integration: the log-likelihood against a closed form, analytic gradients against finite differences, replayed against simulated log-likelihoods, the statistics of the new scheduler against the previous one, and parameter recovery from simulated data (Secs. III and IV). Every number, table and figure in the paper is produced by scripts from the code and the public data, so that it can be regenerated and checked. The author takes full responsibility for the content of the paper.

ACKNOWLEDGMENTS

[To be completed by the author.]

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